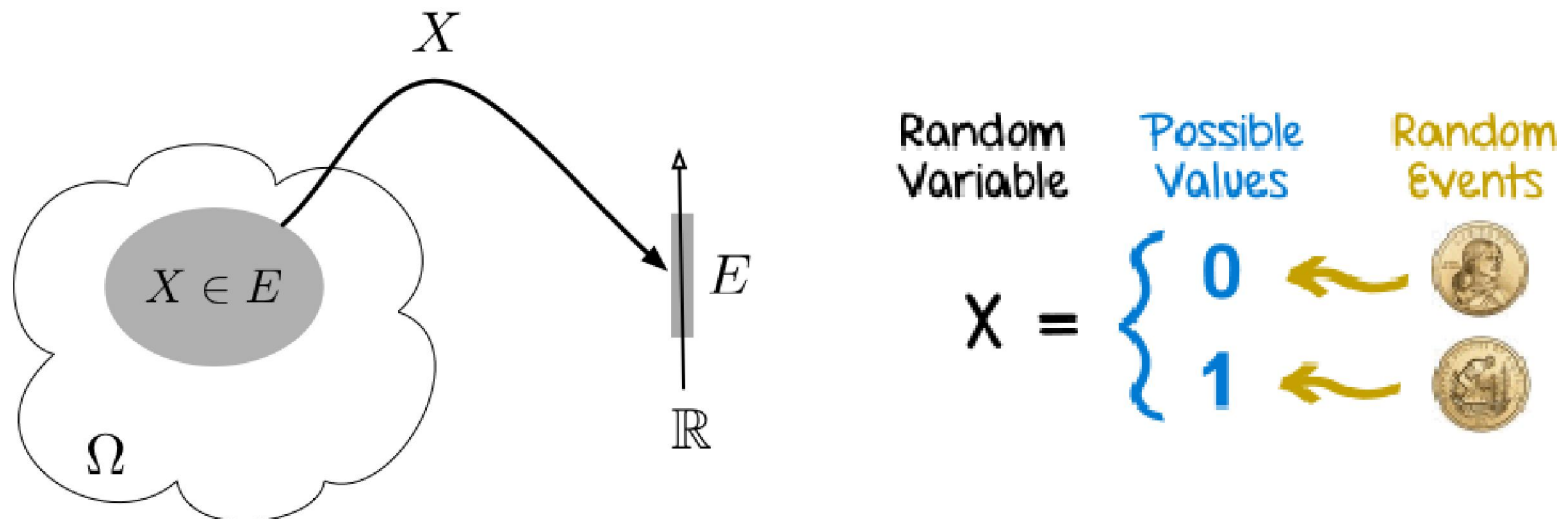


Probability Distributions

Da Zhou (周达)
zhouda@xmu.edu.cn

Random Variable

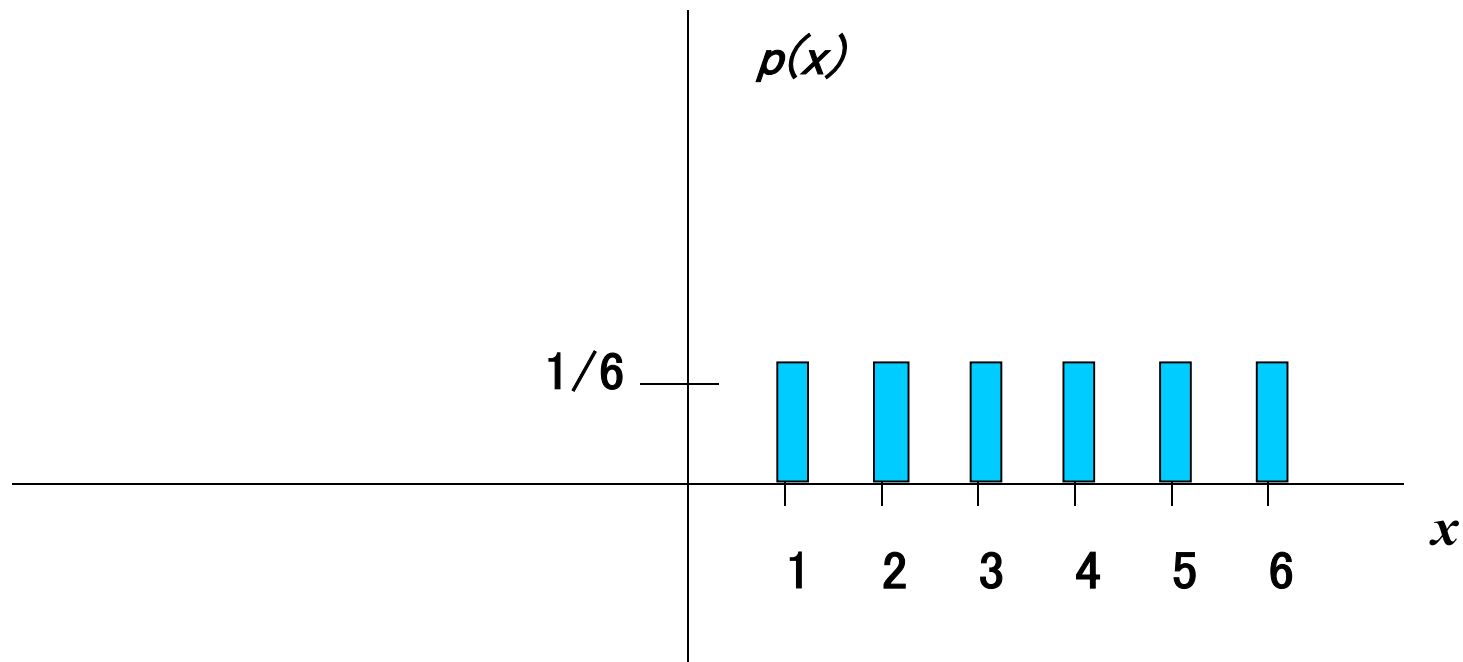
- A random variable X takes on a defined set of values with different probabilities.
 - For example, if you roll a die, the outcome is random and there are 6 possible outcomes, each of which occur with probability one-sixth.



Random variables can be discrete or continuous

- **Discrete** random variables have a countable number of outcomes
 - Examples: Dead/alive, treatment/placebo, dice, counts, etc.
- **Continuous** random variables have an infinite continuum of possible values.
 - Examples: blood pressure, weight, the speed of a car, the real numbers from 1 to 6.

Discrete example: roll of a die



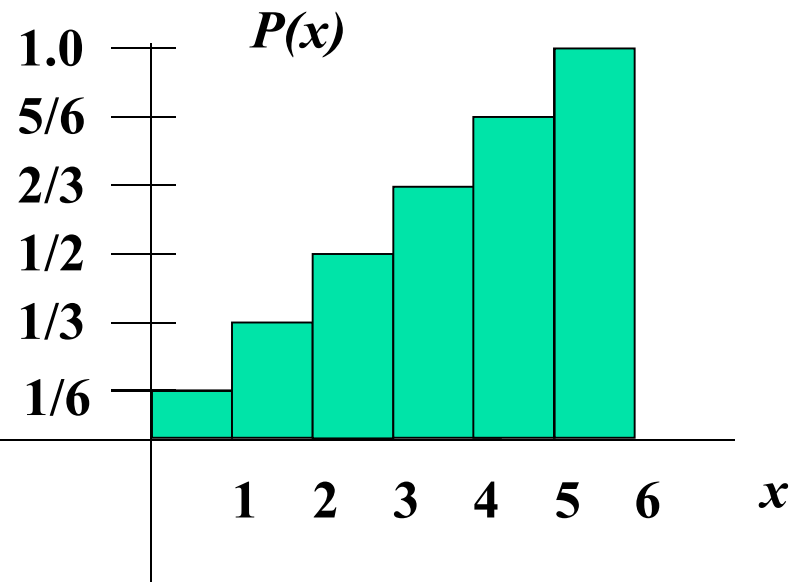
$$\sum_{\text{all } x} P(x) = 1$$

Probability mass function

x	$p(x)$
1	$p(x=1)=1/6$
2	$p(x=2)=1/6$
3	$p(x=3)=1/6$
4	$p(x=4)=1/6$
5	$p(x=5)=1/6$
6	<u>$p(x=6)=1/6$</u>
1.0	

Cumulative distribution function (CDF)

x	$P(X \leq x)$
1	$P(x \leq 1) = 1/6$
2	$P(x \leq 2) = 2/6$
3	$P(x \leq 3) = 3/6$
4	$P(x \leq 4) = 4/6$
5	$P(x \leq 5) = 5/6$
6	$P(x \leq 6) = 6/6$



Important discrete distributions

- Binomial

- Yes/no outcomes (dead/alive, treated/untreated, smoker/non-smoker, sick/well, etc.)

- Poisson

- Counts (e.g., how many cases of disease in a given area)

Continuous case

- The probability function that accompanies a continuous random variable is a continuous mathematical function that integrates to 1.
- The probabilities associated with continuous functions are just areas under the curve (integrals!).
- Probabilities are given for a range of values, rather than a particular value (e.g., the probability of getting a math score between 70 and 80 is 50%).

Continuous case

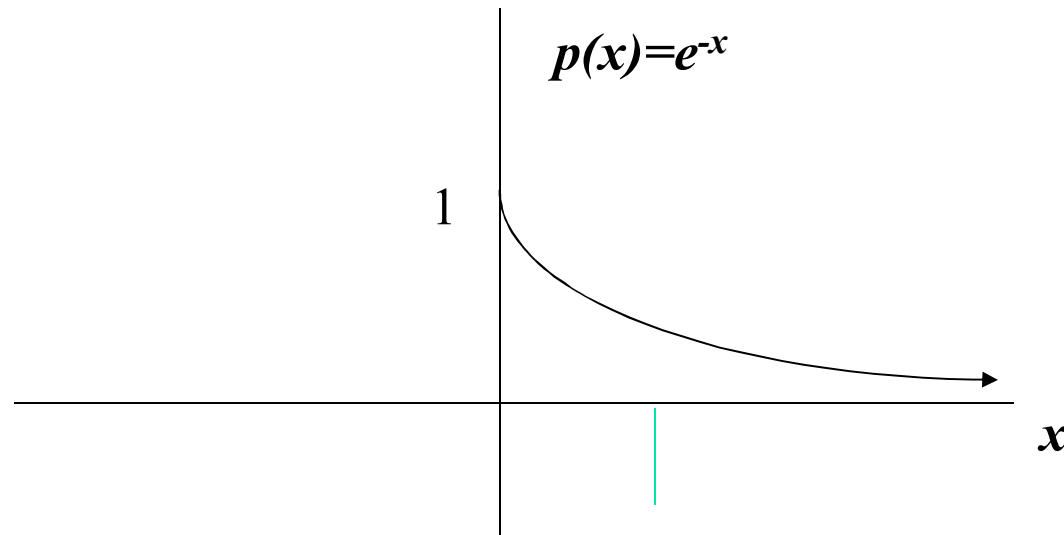
For example, recall the negative exponential function (in probability, this is called an “exponential distribution”):

$$f(x) = e^{-x}$$

This function integrates to 1:

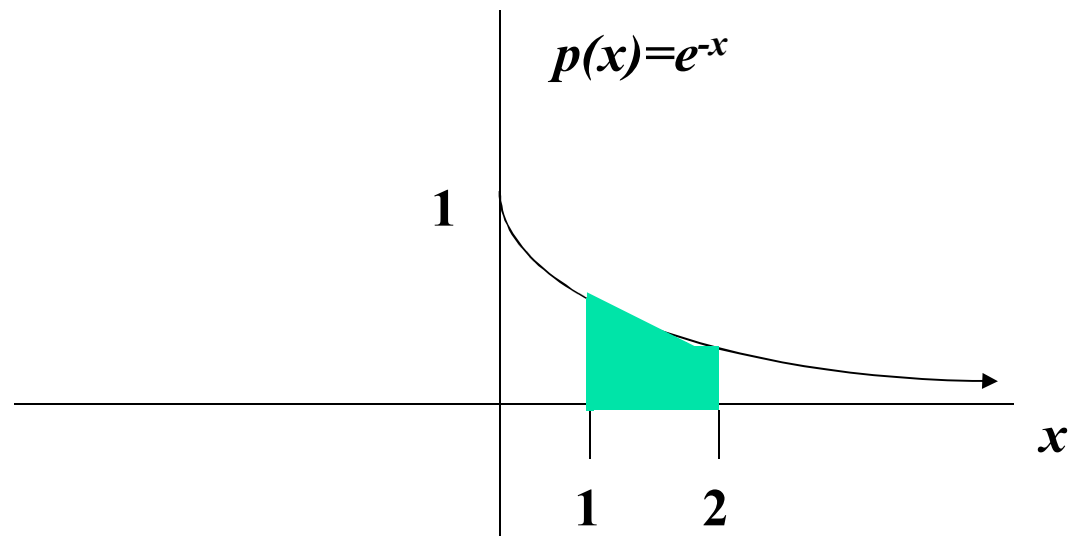
$$\int_0^{+\infty} e^{-x} = -e^{-x} \Big|_0^{+\infty} = 0 + 1 = 1$$

Probability density function (pdf)



The probability that x is any exact particular value (such as 1.9976) is 0; we can only assign probabilities to possible ranges of x .

For example, the probability of x falling within 1 to 2:



$$P(1 \leq X \leq 2) = \int_1^2 e^{-x} = -e^{-x} \Big|_1^2 = -e^{-2} - (-e^{-1}) = -.135 + .368 = .23$$

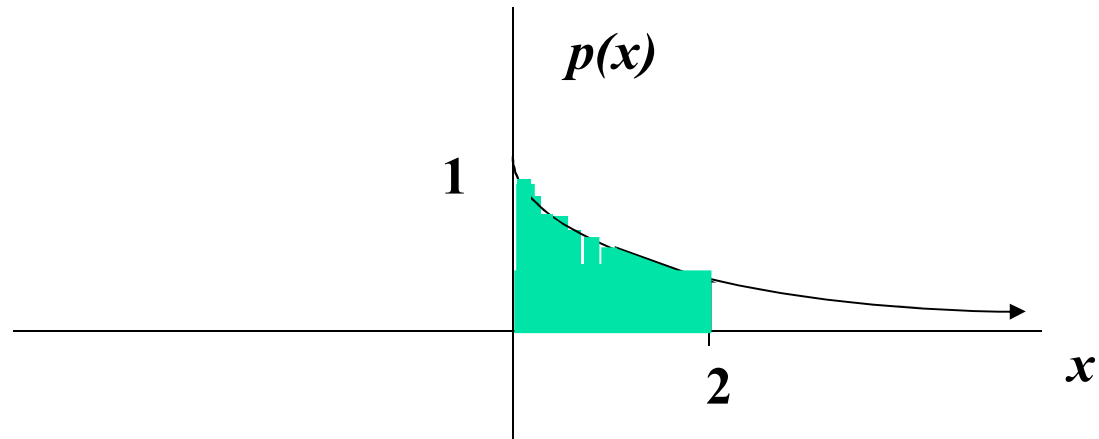
Cumulative distribution function

As in the discrete case, we can specify the “cumulative distribution function” (CDF):

The CDF here = $P(X \leq A) =$

$$\int_0^A e^{-x} = -e^{-x} \Big|_0^A = -e^{-A} - (-e^0) = -e^{-A} + 1 = 1 - e^{-A}$$

Example



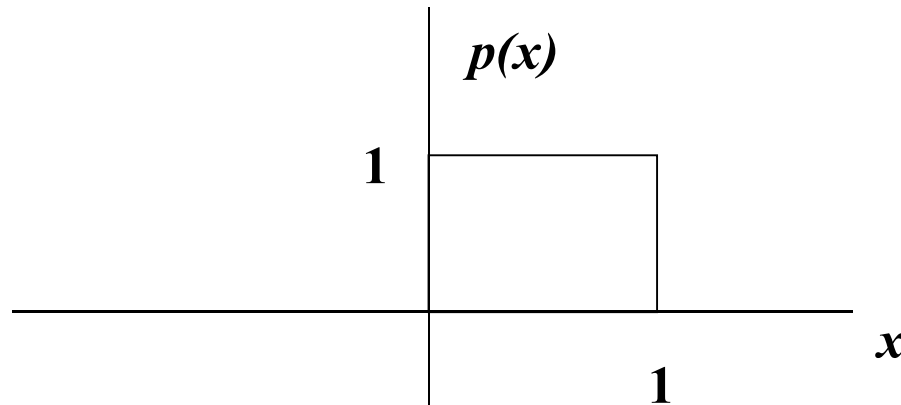
$$P(X \leq 2) = 1 - e^{-2} = 1 - .135 = .865$$

Example 2: Uniform distribution

The uniform distribution: all values are equally likely

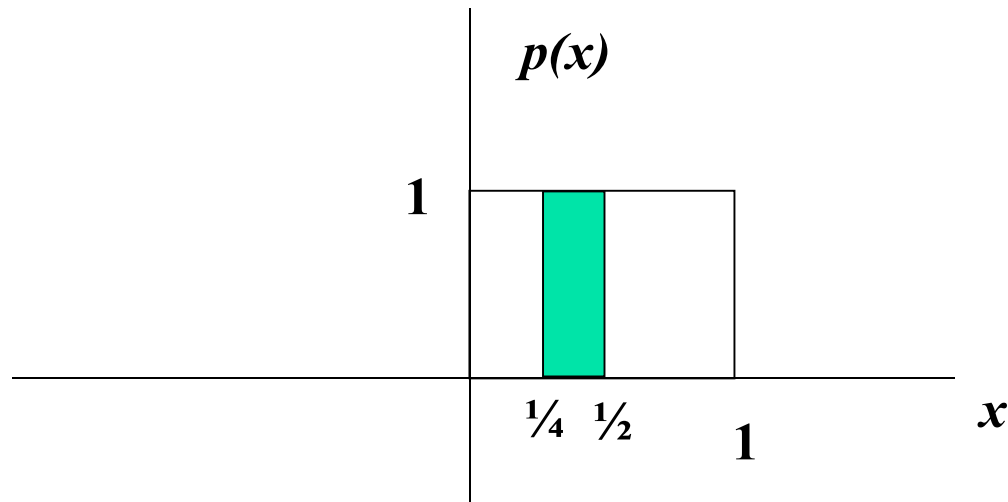
The uniform distribution in $[0,1]$:

$$p(x) = 1, \text{ for } 1 \geq x \geq 0$$



Example: Uniform distribution

What's the probability that x is between $\frac{1}{4}$ and $\frac{1}{2}$?



The relation between CDF and PDF

$$F(x) = \int_{-\infty}^x p(t) dt$$

$$p(x) = F'(x)$$

Expectation, expected value (or mean value)

- If we understand the underlying probability function of a certain phenomenon, then we can make informed decisions based on how we expect X to behave on-average over the long-run...(so called “frequentist” theory of probability).
- Expected value is just the weighted average or mean (μ) of random variable X .

- Given the following probability distribution:

x	10	11	12	13	14
$P(x)$	.4	.2	.2	.1	.1

$$\sum_{i=1}^5 x_i p(x) = 10(.4) + 11(.2) + 12(.2) + 13(.1) + 14(.1) = 11.3$$

Expected value, formally

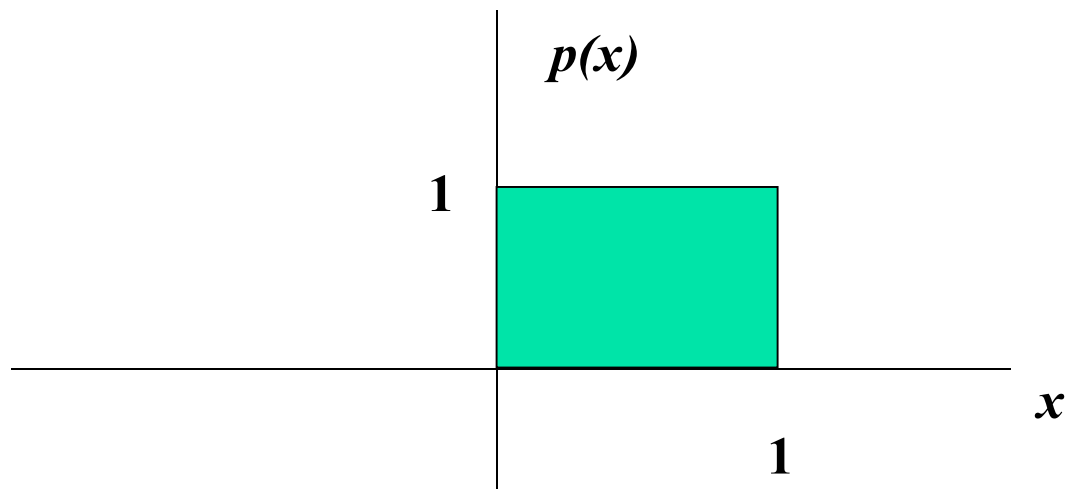
Discrete case:

$$\mu = E(X) = \sum_{\text{all } x} x_i p(x_i)$$

Continuous case:

$$\mu = E(X) = \int_{-\infty}^{+\infty} x p(x) dx$$

Example: uniform distribution



$$E(X) = \int_0^1 1 dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2} - 0 = \frac{1}{2}$$

Example: the lottery

- The Lottery

A certain lottery works by picking 6 numbers from 1 to 49. It costs \$1.00 to play the lottery, and if you win (the magic number is picked by you), you win \$2 million after taxes.

If you play the lottery once, what are your expected winnings or losses?

Lottery

Calculate the probability of winning in 1 try:

$$\frac{1}{\binom{49}{6}} = \frac{1}{\frac{49!}{43!6!}} = \frac{1}{13,983,816} = 7.2 \times 10^{-8}$$

The probability function (note, sums to 1.0):

$x\$$	$p(x)$
-1	.999999928
+ 2 million	7.2×10^{-8}

Expected Value

The probability function

x	$p(x)$
-1	.999999928
+ 2 million	7.2×10^{-8}

Expected Value

$$\begin{aligned} E(X) &= P(\text{win}) * \$2,000,000 + P(\text{lose}) * -\$1.00 \\ &= 2.0 \times 10^6 * 7.2 \times 10^{-8} + .999999928 (-1) = .144 - .999999928 = -\$.86 \end{aligned}$$

A few notes about Expected Value

If c = a constant number (i.e., not a variable) and X and Y are any random variables...

- $E(c) = c$
- $E(cX) = cE(X)$
- $E(c + X) = c + E(X)$

Expected value doesn't help...

$x\$$	$p(x)$
0	.50
+\$400,000	.50

$$\mu = E(X) = \sum_{\text{all } x} x_i p(x_i) = 0(.50) + 400,000(.50) = 200,000$$

$x\$$	$p(x)$
+\$200,000	1.0

$$\mu = E(X) = 200,000$$

Variance/standard deviation

“The average (expected) squared distance (or deviation) from the mean”

$$\sigma^2 = Var(x) = E[(x - \mu)^2] = \sum_{\text{all } x} (x_i - \mu)^2 p(x_i)$$

Variance, formally

Discrete case:

$$Var(X) = \sigma^2 = \sum_{\text{all } x} (x_i - \mu)^2 p(x_i)$$

Continuous case:

$$Var(X) = \sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx$$

Variance: Deal or No Deal

$$\sigma^2 = \sum_{\text{all } x} (x_i - \mu)^2 p(x_i)$$

$$\begin{aligned}\sigma^2 &= \sum_{\text{all } x} (x_i - \mu)^2 p(x_i) = \\ &= (1 - 200,000)^2 (.5) + (400,000 - 200,000)^2 (.5) = 200,000^2 \\ \sigma &= \sqrt{200,000^2} = 200,000\end{aligned}$$

Handy calculation formula!

$$\text{Var}(x) = E(x - \mu)^2 = E(x^2) - [E(x)]^2$$

A few notes about Variance

If c = a constant number (i.e., not a variable) and X and Y are random variables, then

- $\text{Var}(c) = 0$
- $\text{Var}(c+X) = \text{Var}(X)$
- $\text{Var}(cX) = c^2 \text{Var}(X)$

Examples of discrete probability distributions

Binomial Probability Distribution

- A fixed number of observations (trials), n
 - e.g., 15 tosses of a coin; 20 patients; 1000 people surveyed
- A binary random variable
 - e.g., head or tail in each toss of a coin; defective or not defective light bulb
 - Generally called “success” and “failure”
 - Probability of success is p , probability of failure is $1 - p$
- Constant probability for each observation
 - e.g., Probability of getting a tail is the same each time we toss the coin

Example

Take the example of 5 coin tosses. What's the probability that you flip exactly 3 heads in 5 coin tosses?

Binomial distribution

Solution:

One way to get exactly 3 heads: HHHTT

What's the probability of this exact arrangement?

$$\begin{aligned} &P(\text{heads}) \times P(\text{heads}) \times P(\text{heads}) \times P(\text{tails}) \times P(\text{tails}) \\ &= (1/2)^3 \times (1/2)^2 \end{aligned}$$

Another way to get exactly 3 heads: THHHT

$$\begin{aligned} \text{Probability of this exact outcome} &= (1/2)^1 \times (1/2)^3 \\ &\times (1/2)^1 = (1/2)^3 \times (1/2)^2 \end{aligned}$$

Binomial distribution

In fact, $(1/2)^3 \times (1/2)^2$ is the probability of each unique outcome that has exactly 3 heads and 2 tails.

So, the overall probability of 3 heads and 2 tails is:

$(1/2)^3 \times (1/2)^2 + (1/2)^3 \times (1/2)^2 + (1/2)^3 \times (1/2)^2$
+ for as many unique arrangements as there are—but how many are there??

$$\binom{5}{3}$$

ways to
arrange 3
heads in
5 trials

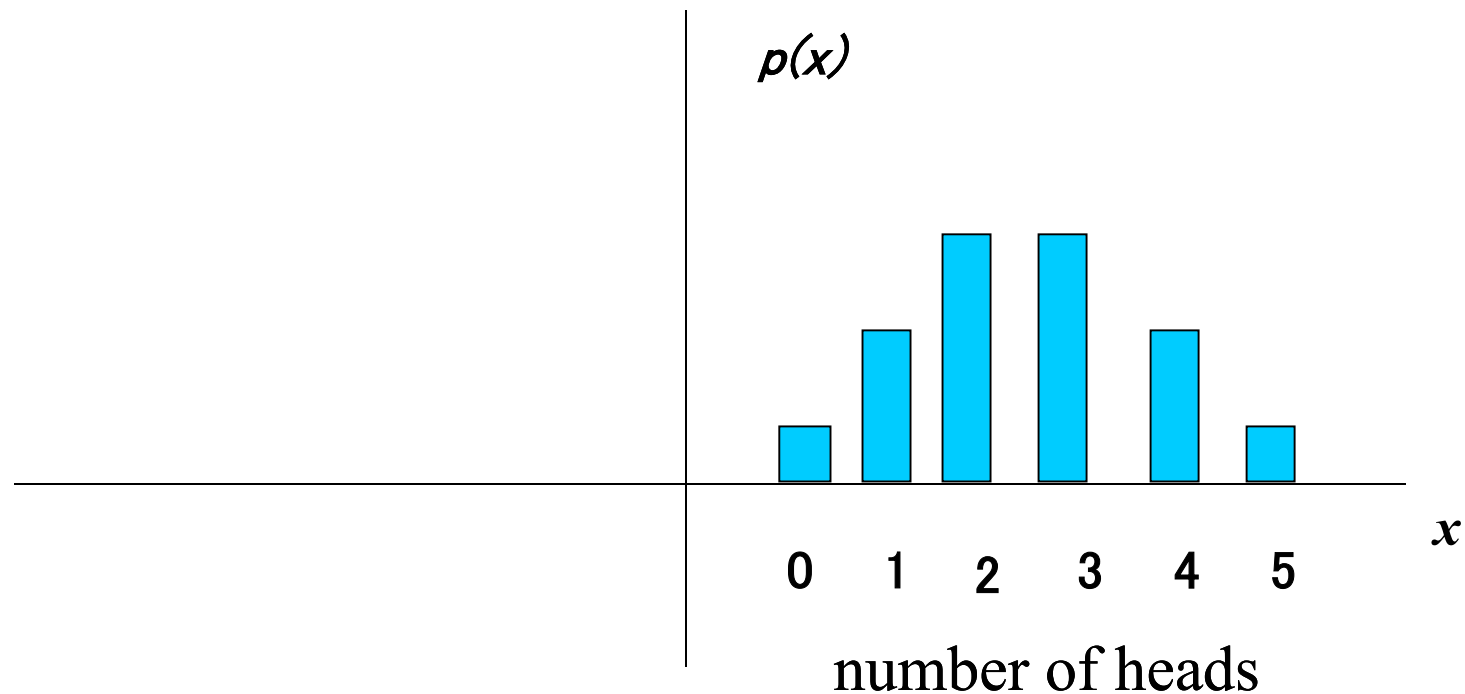
$$= 5!/3!2! = 10$$

Outcome	Probability
THHHT	$(1/2)^3 \times (1/2)^2$
HHHTT	$(1/2)^3 \times (1/2)^2$
TTHHH	$(1/2)^3 \times (1/2)^2$
HTTHH	$(1/2)^3 \times (1/2)^2$
HHTTH	$(1/2)^3 \times (1/2)^2$
THTHH	$(1/2)^3 \times (1/2)^2$
HTHTH	$(1/2)^3 \times (1/2)^2$
HHTHT	$(1/2)^3 \times (1/2)^2$
THHTH	$(1/2)^3 \times (1/2)^2$
HTHHT	$(1/2)^3 \times (1/2)^2$
10 arrangements $\times (1/2)^3 \times (1/2)^2$	

The probability
of each unique
outcome (note:
they are all equal)

Binomial distribution function:

X = the number of heads tossed in 5 coin
tosses



Note the general pattern emerging → if you have only two possible outcomes (call them 1/0 or yes/no or success/failure) in n independent trials, then the probability of exactly X “successes”=

The diagram shows the binomial probability formula $\binom{n}{X} p^X (1-p)^{n-X}$ enclosed in a purple rectangular box. Four arrows point from descriptive text to parts of the formula: one from ' n ' to ' $n = \text{number of trials}$ ', one from ' X ' to ' $X = \# \text{ successes out of } n \text{ trials}$ ', one from ' p ' to ' $p = \text{probability of success}$ ', and one from ' $1-p$ ' to ' $1-p = \text{probability of failure}$ '.

$$\binom{n}{X} p^X (1-p)^{n-X}$$

$n = \text{number of trials}$

$X = \#$
successes
out of n
trials

$p =$
probability of
success

$1-p = \text{probability}$
of failure

Definitions: Binomial

- **Binomial:** Suppose that n independent experiments, or trials, are performed, where n is a fixed number, and that each experiment results in a “success” with probability p and a “failure” with probability $1-p$. The total number of successes, X , is a binomial random variable with parameters n and p .
- We write: **$X \sim \text{Bin}(n, p)$** {reads: “ X is distributed binomially with parameters n and p ”}
- And the probability that $X=r$ (i.e., that there are exactly r successes) is:

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

Bernoulli trial: If there is only 1 trial with probability of success p and probability of failure $1-p$, this is called a Bernoulli distribution. (special case of the binomial with $n=1$)

Probability of success:

$$P(X = 1) = \binom{1}{1} p^1 (1-p)^{1-1} = p$$

Probability of failure:

$$P(X = 0) = \binom{1}{0} p^0 (1-p)^{1-0} = 1-p$$

If X follows a binomial distribution with parameters n and p : $\mathbf{X \sim Bin (n, p)}$

Then:

$$\mu_x = E(X) = np$$

$$\sigma_x^2 = \text{Var}(X) = np(1-p)$$

$$\sigma_x = \text{SD}(X) = \sqrt{np(1-p)}$$

Note: $p(1-p)$ reaches maximum at $p=0.5$
 $p(1-p)=0.25$

For Bernoulli ($n=1$)

$$E(X) = p \quad \text{Var}(X) = p(1-p)$$

Practice problems

- If the probability of developing disease in the exposed group is .05 for the study duration, then if you sample (randomly) 500 exposed people, how many do you expect to develop the disease? What's the probability that **at most** 10 exposed people develop the disease?

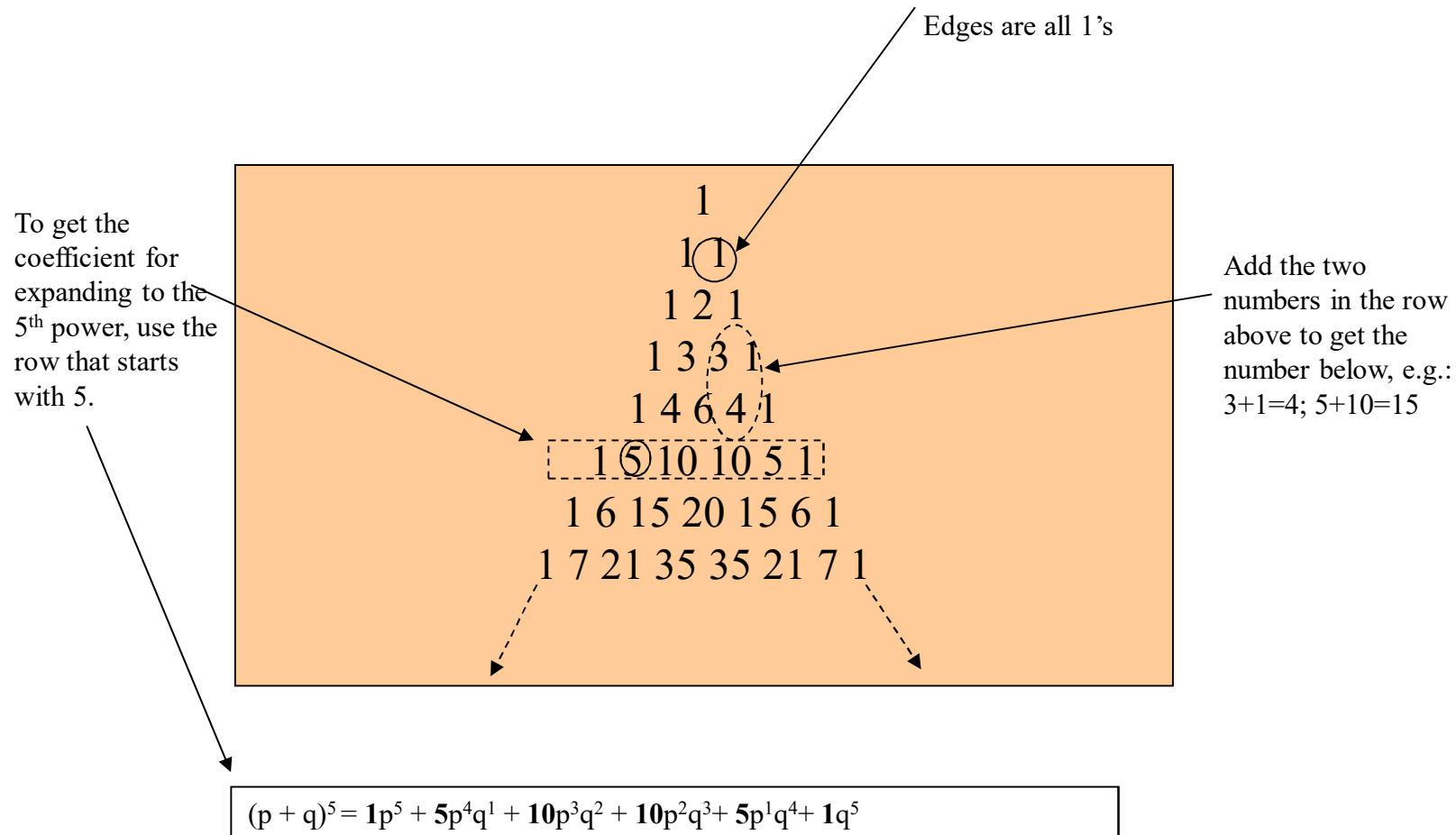
Answer

This is asking for a CUMULATIVE PROBABILITY: the probability of 0 getting the disease or 1 or 2 or 3 or 4 or up to 10.

$$P(X \leq 10) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + \dots + P(X=10) =$$

$$\binom{500}{0} (.05)^0 (.95)^{500} + \binom{500}{1} (.05)^1 (.95)^{499} + \binom{500}{2} (.05)^2 (.95)^{498} + \dots + \binom{500}{10} (.05)^{10} (.95)^{490} < .01$$

Pascal's Triangle



Relationship between binomial probability distribution and binomial expansion

If $p + q = 1$ (which is the case if they are binomial probabilities)

then: $(p + q)^5 = (1)^5 = 1$ or, equivalently:

$$1p^5 + 5p^4q^1 + 10p^3q^2 + 10p^2q^3 + 5p^1q^4 + 1q^5 = 1$$

(the probabilities sum to 1, making it a probability distribution!)

$P(X=0)$ $P(X=1)$ $P(X=2)$ $P(X=3)$ $P(X=4)$ $P(X=5)$

Multinomial distribution

The multinomial is a generalization of the binomial. It is used when there are more than 2 possible outcomes

- Instead of partitioning n trials into 2 outcomes (yes with probability p / no with probability $1-p$), you are partitioning n trials into 3 or more outcomes (with probabilities: $p_1, p_2, p_3, \dots$)
 - General formula for 3 outcomes:

$$P(D = x, R = y, G = z) = \frac{n!}{x! y! z!} p_D^x p_R^y (1 - p_D - p_R)^z$$

Multinomial example

if you are randomly choosing 8 people from an audience that contains 50% democrats, 30% republicans, and 20% green party, what's the probability of choosing exactly 4 democrats, 3 republicans, and 1 green party member?

$$P(D = 4, R = 3, G = 1) = \frac{8!}{4!3!1!} (.5)^4 (.3)^3 (.2)^1$$

Poisson Distribution

- Poisson distribution is for counts—if events happen at a constant rate over time, the Poisson distribution gives the probability of X number of events occurring in time T .
- $X \sim \text{Poisson}(\lambda)$, then the probability that $X=k$ (a particular count) is:

$$p(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

Poisson Mean and Variance

- Mean $\mu = \lambda$
- Variance $\sigma^2 = \lambda$

where λ = expected number of hits in a given time period

Poisson process

Note that the Poisson parameter λ can be given as the mean number of events that occur in a defined time period OR, equivalently, λ can be given as a rate, such as $\lambda=2/\text{month}$ (2 events per 1 month) that must be multiplied by $t=\text{time}$ (called a "Poisson Process")

$$P(X(t)-X(0) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

Example

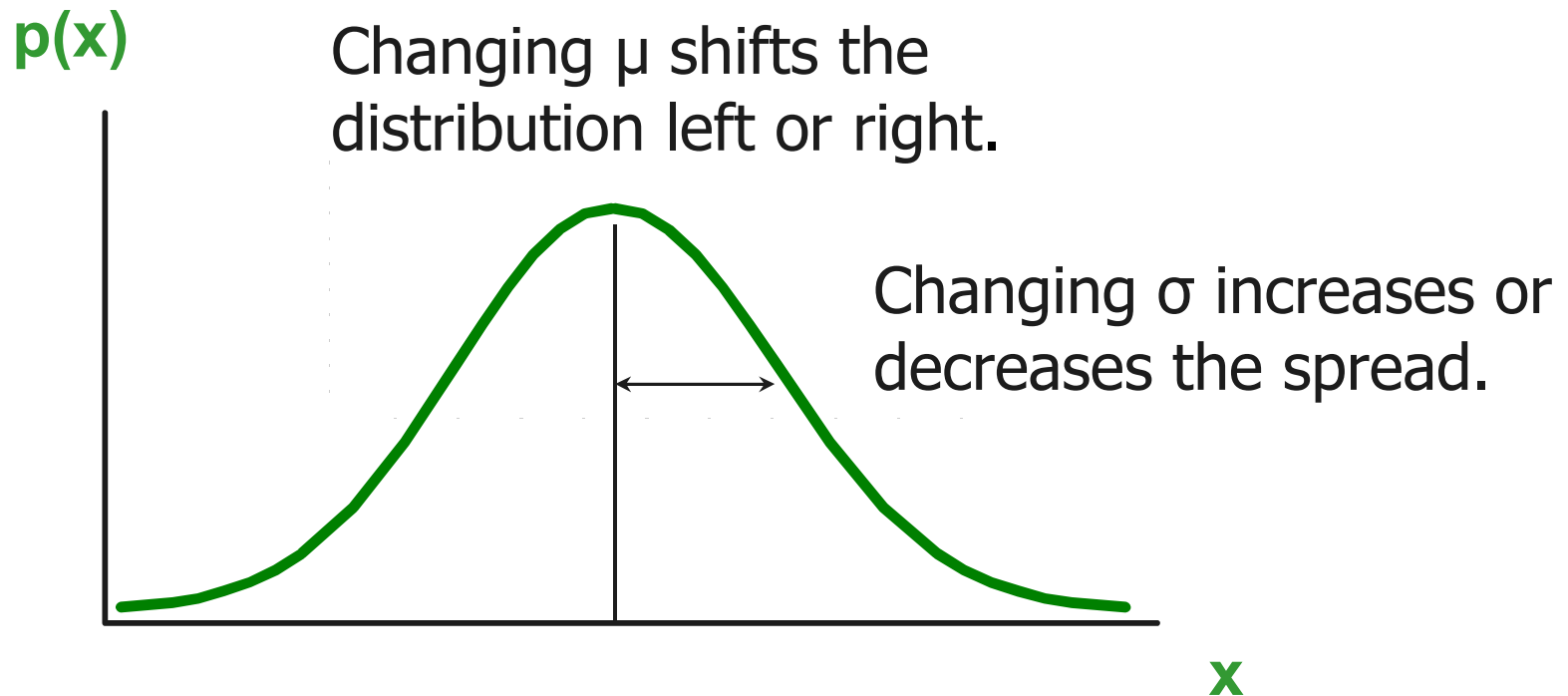
For example, if new cases of disease in some city are occurring at a rate of about 2 per day, then what's the probability that exactly 4 cases will occur in the next 3 days?

$X \sim \text{Poisson } (\lambda=2/\text{day})$

$$P(X(3)-X(0) = 4) = \frac{(2*3)^4 e^{-(2*3)}}{4!} = \frac{6^4 e^{-(6)}}{4!} = 13.4\%$$

Examples of continuous probability distributions

The Normal Distribution



Definition $N(\mu, \sigma^2)$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

This is a bell shaped curve with different centers and spreads depending on μ and σ

The Normal PDF

It's a probability function, so no matter what the values of μ and σ , must integrate to 1!

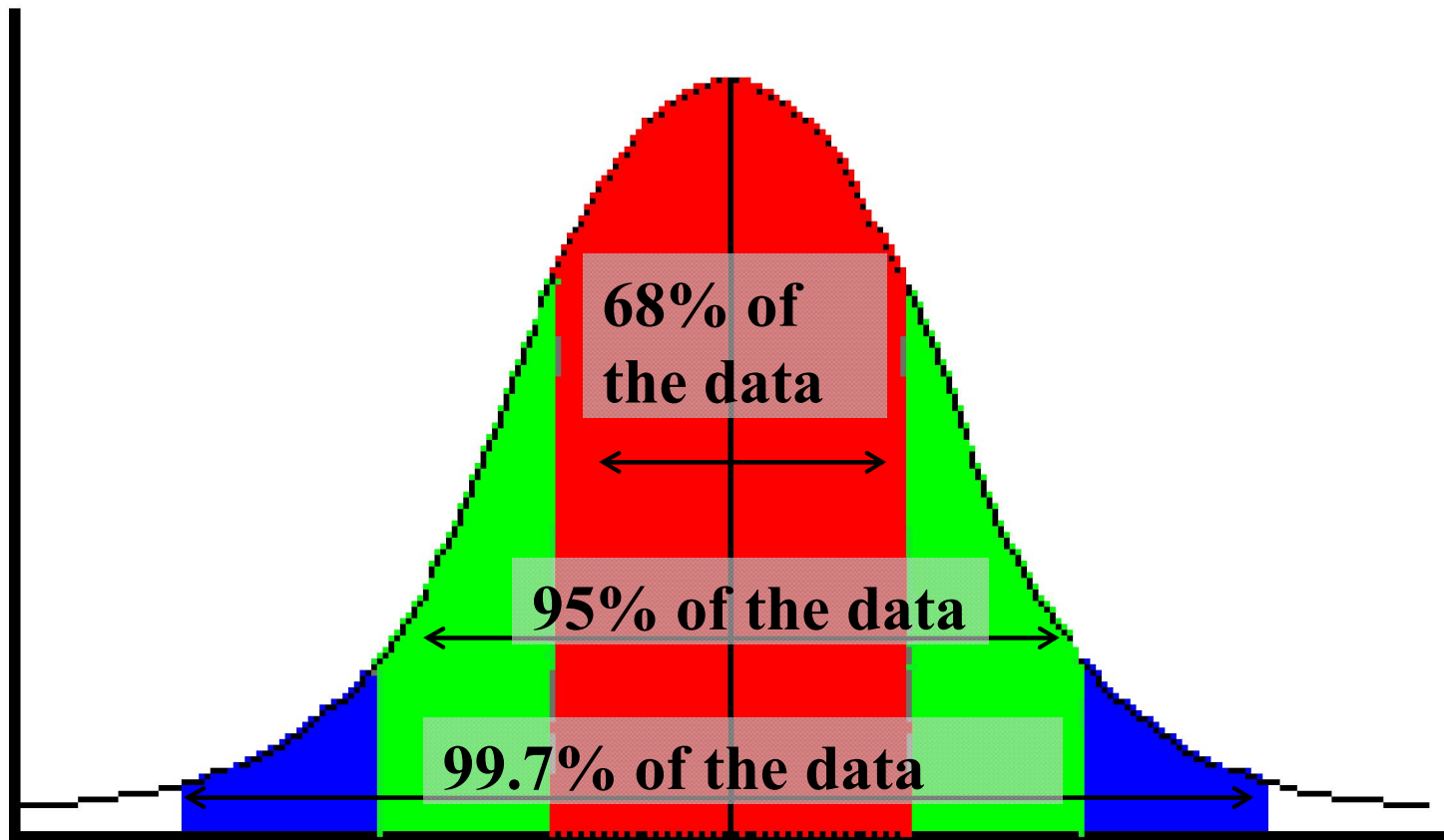
$$\int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 1$$

$$E(X)=\mu = \int_{-\infty}^{+\infty} x \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

$$\text{Var}(X)=\sigma^2 = \int_{-\infty}^{+\infty} x^2 \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx - \mu^2$$

$$\text{Standard Deviation}(X)=\sigma$$

68-95-99.7 Rule



Example

- Suppose SAT scores roughly follows a normal distribution in the U.S. population of college-bound students (with range restricted to 200-800), and the average math SAT is 500 with a standard deviation of 50, then:
 - 68% of students will have scores between 450 and 550
 - 95% will be between 400 and 600
 - 99.7% will be between 350 and 650

Example

- BUT...
- What if you wanted to know the math SAT score corresponding to the 90th percentile (=90% of students are lower)?

$$P(X \leq Q) = .90 \rightarrow$$

$$\int_{200}^Q \frac{1}{(50)\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-500}{50}\right)^2} dx = .90$$

Solve for Q?

The Standard Normal $N(0,1)$

The formula for the standardized normal probability density function is

$$p(Z) = \frac{1}{(1)\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{Z-0}{1}\right)^2} = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}(Z)^2}$$

Normalization

All normal distributions can be converted into the standard normal curve by subtracting the mean and dividing by the standard deviation:

$$Z = \frac{X - \mu}{\sigma}$$

Example

- For example: What's the probability of getting a math SAT score of 575 or less, $\mu=500$ and $\sigma=50$?

$$Z = \frac{575 - 500}{50} = 1.5$$

- i.e., A score of 575 is 1.5 standard deviations above the mean

$$\therefore P(X \leq 575) = \int_{200}^{575} \frac{1}{(50)\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-500}{50}\right)^2} dx \longrightarrow \int_{-\infty}^{1.5} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}z^2} dz$$

Look up $Z=1.5$ in standard normal chart (or enter into SAS) $\rightarrow 0.9332$

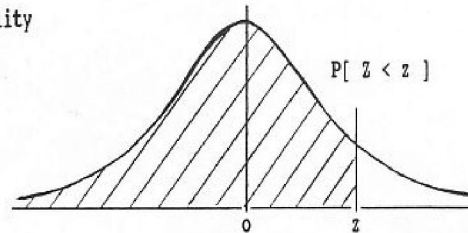
Looking up probabilities in the standard normal table

STANDARD STATISTICAL TABLES

1. Areas under the Normal Distribution

The table gives the cumulative probability up to the standardised normal value z i.e.

$$P[Z < z] = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) dz$$



What is the area to the left of $Z=1.51$ in a standard normal curve?

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5159	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7854
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8804	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

$Z=1.51$

$Z=1.51$

Area is 93.45%

Are my data “normal”?

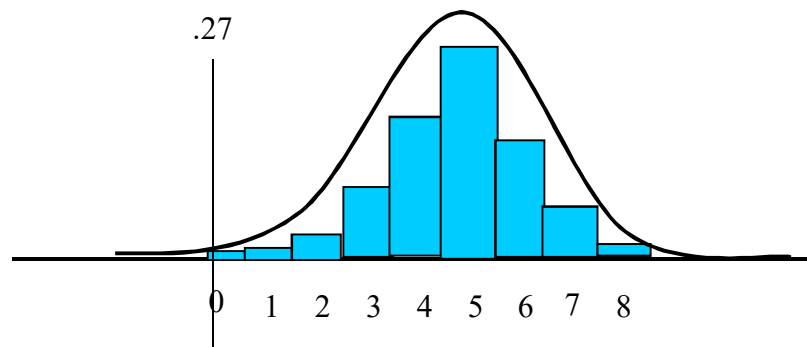
- Not all continuous random variables are normally distributed!!
- It is important to evaluate how well the data are approximated by a normal distribution

Are my data normally distributed?

1. Look at the histogram! Does it appear bell shaped?
2. Compute descriptive summary measures—are mean, median, and mode similar?
3. Do 2/3 of observations lie within 1 std dev of the mean? Do 95% of observations lie within 2 std dev of the mean?
4. Look at a normal probability plot—is it approximately linear?
5. Run tests of normality (such as Kolmogorov-Smirnov). But, be cautious, highly influenced by sample size!

Normal approximation to binomial

When you have a binomial distribution where n is large and p isn't too small (rule of thumb: $\text{mean} > 5$), then the binomial starts to look like a normal distribution →



Practice problem

If the probability of developing disease in the exposed group is .25, then if you sample (randomly) 500 exposed people, What's the probability that **at most** 120 people develop the disease?

Answer

By hand:

$$P(X \leq 120) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + \dots + P(X=120) = \\ \binom{500}{120} (.25)^{120} (.75)^{380} + \binom{500}{2} (.25)^2 (.75)^{498} + \binom{500}{1} (.25)^1 (.75)^{499} + \binom{500}{0} (.25)^0 (.75)^{500} \dots$$

OR use, normal approximation:

$$\mu = np = 500(.25) = 125 \text{ and } \sigma^2 = np(1-p) = 93.75; \sigma = 9.68$$

$$Z = \frac{120 - 125}{9.68} = -.52$$

$$P(Z < -.52) = .3015$$